

23-4b The GDP Deflator

As we have just seen, nominal GDP reflects both the quantities of goods and services the economy is producing and the prices of those goods and services. By contrast, by holding prices constant at base-year levels, real GDP reflects only the quantities produced. From these two statistics, we can compute a third, called the GDP deflator, which reflects only the prices of goods and services.

The

GDP deflator (a measure of the price level calculated as the ratio of nominal GDP to real GDP times 100)

is calculated as follows:

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100.$$

Because nominal GDP and real GDP must be the same in the base year, the GDP deflator for the base year always equals 100. The GDP deflator for subsequent years measures the change in nominal GDP from the base year that cannot be attributable to a change in real GDP.

The GDP deflator measures the current level of prices relative to the level of prices in the base year. To see why this is true, consider a couple of simple examples. First, imagine that the quantities produced in the economy rise over time but prices remain the same. In this case, both nominal and real GDP rise at the same rate, so the GDP deflator is constant. Now suppose, instead, that prices rise over time but the quantities produced stay the same. In this second case, nominal GDP rises but real GDP remains the same, so the GDP deflator rises. Notice that, in both cases, the GDP deflator reflects what's happening to prices but not to quantities.

Let's now return to our numerical example in [Table 2](#). The GDP deflator is computed at the bottom of the table. For the year 2019, nominal GDP is \$200 and real GDP is \$200, so the GDP deflator is 100. (The deflator is always 100 in the base year.) For the year 2020, nominal GDP is \$600 and real GDP is \$350, so the GDP deflator is 171.

Economists use the term *inflation* to describe a situation in which the economy's overall price level is rising. The *inflation rate* is the percentage change in some measure of the price level from one period to the next. Using the GDP deflator, the inflation rate between two consecutive years is computed as follows:

$$\text{Inflation rate in year 2} = \frac{\text{GDP deflator in year 2} - \text{GDP deflator in year 1}}{\text{GDP deflator in year 1}} \times 100.$$

Because the GDP deflator rose in year 2020 from 100 percent to 171 percent, the inflation rate is $100 \times (171 - 100)/100$, or 71 percent. In 2021, the GDP deflator rose to 240 from 171 the previous year, so the inflation rate is $100 \times (240 - 171)/171$, or 40 percent.

The GDP deflator is one measure that economists use to monitor the average level of prices in the economy and thus the rate of inflation. The GDP deflator gets its name because it can be used to take inflation out of nominal GDP—that is, to “deflate” nominal GDP for the rise that is due to increases in prices. In the next chapter, we examine another measure of the economy’s price level, called the *consumer price index*, and discuss the differences between the two measures.

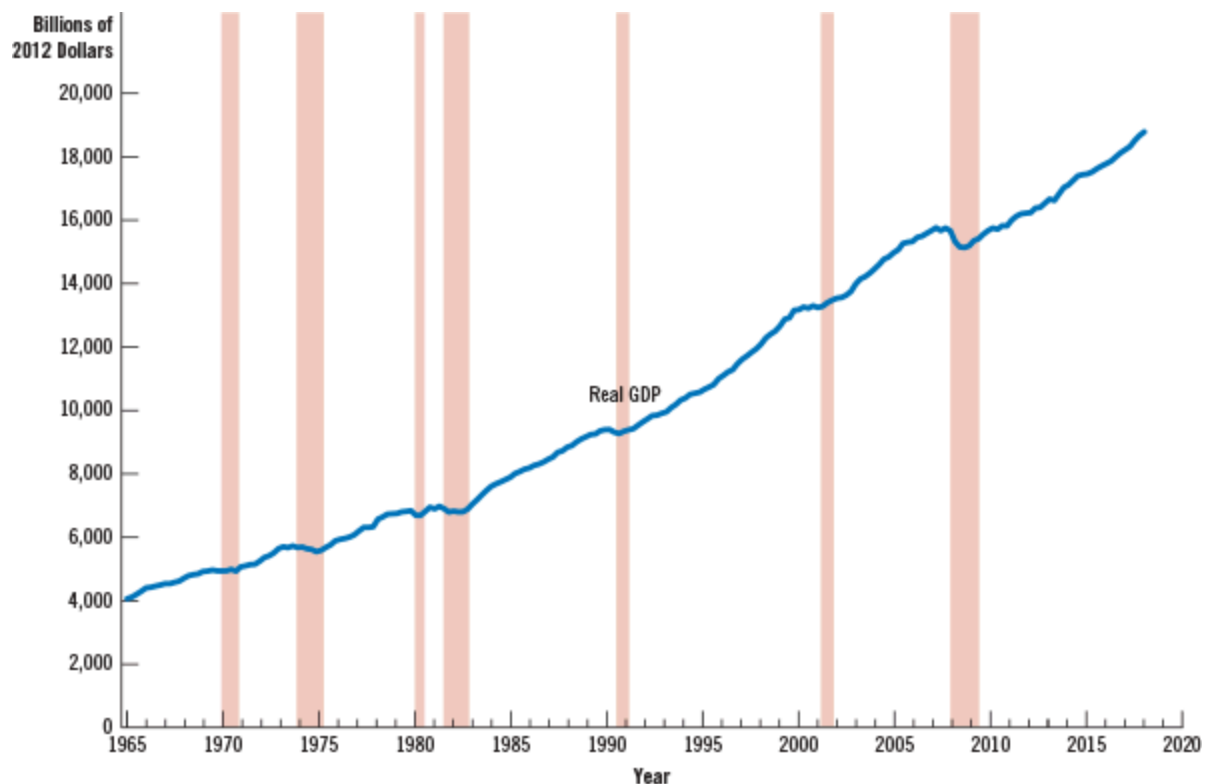
Case Study: A Half Century of Real GDP

Now that we know how real GDP is defined and measured, let’s look at what this macroeconomic variable tells us about the recent history of the United States. [Figure 2](#) shows quarterly data on real GDP for the U.S. economy since 1965.

Figure 2

Real GDP in the United States

This figure shows quarterly data on real GDP for the U.S. economy since 1965. Recessions—periods of falling real GDP—are marked with the shaded vertical bars.



Source: U.S. Department of Commerce.

The most obvious feature of these data is that real GDP grows over time. The real GDP of the U.S. economy in 2018 was more than four times its 1965 level. Put differently, the output

of goods and services produced in the United States has grown on average about **3** percent per year. Because this continued growth in real GDP exceeds the rate of population growth, it enables most Americans to enjoy greater economic prosperity than their parents and grandparents did.

A second feature of the GDP data is that growth is not steady. The upward climb of real GDP is occasionally interrupted by periods during which GDP declines, called *recessions*. [Figure 2](#) marks recessions with shaded vertical bars. (There is no ironclad rule for when the official business cycle dating committee will declare that a recession has occurred, but an old rule of thumb is two consecutive quarters of falling real GDP.) Recessions are associated not only with lower incomes but also with other forms of economic distress: rising unemployment, falling profits, increased bankruptcies, and so on.

Much of macroeconomics aims to explain the long-run growth and short-run fluctuations in real GDP. As we will see in the coming chapters, we need different models for these two purposes. Because the short-run fluctuations represent deviations from the long-run trend, we first examine the behavior of key macroeconomic variables, including real GDP, in the long run. Then in later chapters, we build on this analysis to explain short-run fluctuations.

QuickQuiz

9. An economy produces **10** cookies in year **1** at a price of **\$2** per cookie and **12** cookies in year **2** at a price of **\$3** per cookie. From year **1** to year **2**, real GDP increases by

- a. **20** percent.
- b. **50** percent.
- c. **70** percent.
- d. **80** percent.

10. If all quantities produced rise by **5** percent and all prices fall by **5** percent, which of the following best describes what occurs?

- a. Real GDP rises by **5** percent, while nominal GDP falls by **5** percent.
- b. Real GDP rises by **5** percent, while nominal GDP is unchanged.
- c. Real GDP is unchanged, while nominal GDP rises by **5** percent.
- d. Real GDP is unchanged, while nominal GDP falls by **5** percent.

Chapter 23: Measuring a Nation's Income: 23-4b The GDP Deflator

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